AI-Powered Startup Validator

Vacation Mini Project Report

IdeaProof

AI-Powered Startup Idea Validator

Validate your startup idea before you build it.

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Department	**Bachelor of Computer Applications (BCA)**
Semester	**4th Semester**
Submission	**July 2026**

https://startup-validator-bhoomika.netlify.app/

CERTIFICATE

This is to certify that the mini project entitled

IDEAPROOF — AI-POWERED STARTUP IDEA VALIDATOR

has been successfully developed and submitted by **BHOOMIKA** (Roll No. **24BCA070**), a student of the 4th Semester, Bachelor of Computer Applications (BCA), St. Claret College (Autonomous), Bengaluru.

This mini project was developed independently as part of self-learning during the vacation period. The work is original and has not been submitted elsewhere for any academic examination or award.

St. Claret College (Autonomous), Bengaluru

DECLARATION

I, **BHOOMIKA**, Roll No. **24BCA070**, 4th Semester, BCA, St. Claret College (Autonomous), Bengaluru, declare that the mini project **"IDEAPROOF — AI-POWERED STARTUP IDEA VALIDATOR"** is a record of original work done by me during the vacation period.

I declare that:

- This project was carried out independently, without any formal institutional guidance.
- The work is genuine and has not been copied from any source without acknowledgement.
- This project has not been submitted for any degree or academic title elsewhere.
- All external APIs and third-party libraries used have been duly cited and acknowledged.
- The conclusions drawn are based entirely on my own understanding and research.

Place: Bengaluru

Date: July 2026

Signature of the Student

BHOOMIKA

Roll No: 24BCA070 | 4th Semester, BCA

ACKNOWLEDGEMENT

I thank all those who encouraged and supported me throughout the development of this project.

I am grateful to the developers of **Chart.js**, **jsPDF**, and **Font Awesome** for providing high-quality open-source libraries freely available via CDN, which power the data visualisation, PDF export, and iconography throughout IdeaProof. I also thank the maintainers of the **Wikipedia API**, **GitHub API**, **Hacker News Algolia API**, and **Reddit JSON API** for providing free public endpoints that allow real-world data to be fetched directly in the browser.

I am also grateful to the open-source developer community and educational resources available through MDN Web Docs, Stack Overflow, and YouTube, which guided my understanding of deterministic seeded randomness, API fetching with fallback handling, SVG animation, and CSS variable-based dark mode implementation.

I thank my family for their constant support and encouragement throughout the vacation period.

BHOOMIKA

Roll No: 24BCA070 | 4th Semester, BCA

St. Claret College (Autonomous), Bengaluru

ABSTRACT

Starting a new business is one of the riskiest endeavours an individual can undertake. Research consistently shows that a significant proportion of startups fail within their first two years, often due to insufficient market validation before building the product. Professional startup validation services, investor pitch consultations, and market research reports are expensive and inaccessible to most student entrepreneurs and early-stage founders.

IdeaProof is a fully frontend-only startup idea validation platform built as a college mini-project. Users enter their startup name, industry, idea description, target audience, and region — and the application fetches real-time signals from Wikipedia, GitHub, Hacker News, and Reddit, then generates a comprehensive multi-dimensional analysis report including a validation score, SWOT analysis, competitor profiles, revenue model suggestions, risk assessment, target audience segmentation, trend charts, improvement suggestions, and an interactive AI Startup Advisor chatbot.

Key Features:

- Startup idea input form: name, industry (13 options), description, target audience, and region.
- 7-step animated analysis pipeline fetching real data from 4 public APIs with fallback mock data.
- Validation Score (0–100) computed via deterministic seeded algorithm with industry-specific bonuses.
- Score Breakdown across 5 categories: Market Need, Innovation, Scalability, Competition, Revenue Potential.
- Startup Readiness Meter: 4-stage progression (Idea → MVP → Launch → Scale).
- SWOT Analysis: Strengths, Weaknesses, Opportunities, Threats.
- Competitor Analysis: real competitor cards from Wikipedia/GitHub APIs with search and sort.
- Competitor Comparison Table with similarity scores.
- Revenue Model Suggestions: 7 models with fit scores and a Top Pick recommendation.
- Target Audience segmentation: Primary, Secondary, and Early Adopters.
- Risk Assessment: 4 risk cards with Medium/High/Low severity ratings.
- Trend Analysis: bar and line charts (Chart.js) showing market demand, investor interest, job postings.
- Improvement Suggestions: 10 numbered, actionable startup tips.
- AI Startup Advisor: chatbot with quick-question prompts and keyword-matched intelligent responses.
- History: localStorage persistence of last 20 analyses with restore and delete.

- PDF Export: jsPDF-powered downloadable report.
- Dark mode with CSS variable toggle, persisted to localStorage.
- Fully responsive: breakpoints at 700px and 420px.

Technologies Used:

HTML5, CSS3, Vanilla JavaScript (~997 lines), Chart.js, jsPDF, Font Awesome 6.5, Google Fonts (Inter). APIs: Wikipedia, GitHub, Hacker News Algolia, Reddit JSON. Total: ~1,851 lines, ~77KB. Deployed on Netlify.

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INTRODUCTION

1.1 Background

The startup ecosystem rewards founders who validate their ideas before committing significant time, money, and resources to building a product. Market validation — determining whether a genuine customer problem exists, whether the market is large enough, and whether a viable revenue model is achievable — is the critical first step that separates successful startups from those that fail due to building products nobody wants.

Traditional validation methods include customer interviews, landing page tests, competitor analysis, and consulting market research reports. However, these approaches are time-consuming and often require access to expensive databases, business networks, or professional advisors. A tool that aggregates real-world signals from publicly available APIs and applies structured analytical frameworks can democratise startup validation for student entrepreneurs and early-stage founders.

1.2 About the Project

IdeaProof is a fully frontend-only startup idea validation SPA built with HTML5, CSS3, and Vanilla JavaScript — totalling approximately 1,851 lines of code across three files with no build step, no npm, and no framework. It integrates four public APIs (Wikipedia, GitHub, Hacker News, Reddit) to fetch real competitor and trend data, and applies a deterministic seeded scoring algorithm to produce consistent, unique validation reports for every distinct startup idea.

Live at: **https://startup-validator-bhoomika.netlify.app/** — deployed on Netlify.

1.3 Key Design Philosophy

- **Zero signup / zero cost:** Runs entirely in the browser, no backend or API keys required for core features.
- **Deterministic outputs:** The same startup idea always produces the same score and analysis — achieved via a seeded random number generator (LCG algorithm) seeded from a hash of the input string.
- **Real data with graceful fallback:** APIs are called for competitor and trend data; if they fail (rate limits, CORS), pre-written mock datasets are used automatically.
- **SaaS-quality UI:** Dark mode, smooth animations, PDF export, and responsive design — matching the experience of commercial validation tools.

1.4 Project Scale

File	Lines	Contents
index.html	259	Full SPA markup, form, results sections, modal
style.css	595	All styles, dark mode, responsive, animations
app.js	997	All logic, API calls, generators, renderers, chat
Total	1,851	~77KB uncompressed, 3-file project

OBJECTIVES OF THE PROJECT

The following objectives guided the design and development of IdeaProof:

1. To provide instant, structured startup idea validation: Deliver a comprehensive multi-dimensional analysis report within seconds of form submission, covering scoring, SWOT, competitors, revenue models, risks, audience, and trends.

2. To integrate real-world public API data: Fetch actual competitor and market signal data from Wikipedia, GitHub, Hacker News, and Reddit APIs, making the analysis grounded in real-world information rather than purely synthetic outputs.

3. To implement a deterministic scoring algorithm: Design a score engine using a seeded LCG random number generator so that the same startup idea always produces the same validation score, ensuring reproducibility and preventing arbitrary variation between analyses.

4. To cover all key startup validation dimensions: Address the full startup validation framework: market need, innovation, scalability, competition level, revenue potential, SWOT factors, risk categories, target audience segments, and trend signals.

5. To deliver an interactive AI advisor experience: Provide an AI Startup Advisor chatbot with quick-question prompts and keyword-matched intelligent responses that allow founders to ask follow-up questions about their idea.

6. To enable history, export, and persistence: Save the last 20 analyses to localStorage with restore and delete functionality, and provide a jsPDF-based exportable report for offline sharing.

7. To achieve professional SaaS-quality UI without any framework: Implement dark mode, smooth animations, a 7-step loading sequence, responsive layout, and Chart.js visualisations using only HTML, CSS, and Vanilla JavaScript.

EXISTING SYSTEM

3.1 Current Startup Validation Options

Founders and student entrepreneurs currently rely on the following resources for startup validation, each with significant limitations:

a) Professional Market Research Reports

Reports from firms like Gartner, Forrester, or Statista provide authoritative market data but cost thousands of dollars per report — completely inaccessible to students and early-stage founders without institutional access.

b) Startup Consultant and Incubator Services

Business incubators and startup mentors provide personalised validation advice but require application processes, geographic presence, and often equity stakes. They are not scalable, instant, or available on demand.

c) Generic AI Tools (ChatGPT, Claude)

General-purpose AI chatbots can discuss startup ideas but provide no structured scoring, no real competitor data, no SWOT framework output, no revenue model fit scoring, and no persistent history. Each session starts fresh with no consistency.

d) Lean Canvas and Business Model Canvas Templates

Canvas frameworks are valuable strategic tools but require the founder to fill in all fields manually based on their own research — they provide structure without any automated analysis, scoring, or data fetching.

3.2 Limitations Summary

- No free, instant, browser-based startup validator with real API data integration.
- Existing tools require sign-up, payment, or specialist access.
- General AI tools provide no structured scores, history, or PDF export.
- No tool covers the full validation framework (SWOT + competitors + revenue + risks + trends) in one place.
- Canvas templates are passive — no automated analysis or scoring.

PROPOSED SYSTEM

4.1 Overview

IdeaProof proposes a zero-cost, zero-signup, browser-native startup idea validation platform that combines real API data with deterministic algorithmic analysis to deliver instant, structured, reproducible validation reports across all key startup evaluation dimensions — packaged in a SaaS-quality interface with dark mode, charts, history, and PDF export.

4.2 Advantages

- **Instant & Free:** Complete analysis in seconds with no payment, no account, and no installation.
- **Real API Data:** Competitor and trend data fetched from Wikipedia, GitHub, Hacker News, and Reddit — not purely synthetic.
- **Deterministic Scoring:** Same idea = same score every time, via LCG seeded RNG. Reproducible and trustworthy.
- **Full Validation Coverage:** Scores, SWOT, competitors, revenue models, risks, audience, trends, improvements — all in one session.
- **AI Advisor:** Interactive chatbot with contextual startup-specific responses for follow-up questions.
- **Persistent History:** Last 20 analyses saved to localStorage; restorable and deletable at any time.
- **Exportable PDF:** jsPDF-powered report downloadable for sharing with mentors, investors, or co-founders.

APPLICATION ARCHITECTURE

5.1 File Structure

IdeaProof is a 3-file, no-build SPA with zero npm dependencies:

- **index.html (259 lines):** Full application markup — hero, form, loading overlay, and all 12 results section containers.
- **style.css (595 lines):** Complete stylesheet with CSS Custom Property-based dark mode, all component styles, animations, and responsive breakpoints at 700px and 420px.
- **app.js (997 lines):** All application logic — form handling, 7-step analysis pipeline, 4 API fetchers, 10 generator functions, 13 renderer functions, AI chatbot, history management, and PDF export.

5.2 7-Step Analysis Pipeline

On form submission, IdeaProof runs 7 sequential steps with 400–800ms delays and animated progress indicators (pending → spinning → done):

Step	Label	Functions Called
0	Understanding Idea	generateScore() + generateUSP()
1	Finding Competitors	fetchCompetitors() → Wikipedia + GitHub APIs
2	Market Analysis	generateMarket() + generateTrends() + fetchLiveSignals() → HN + Reddit
3	SWOT Generation	generateSWOT()
4	Risk Assessment	generateRisks()
5	Revenue Strategy	generateRevenue() + generateAudience() + generateCanvas()
6	Final Validation	generateImprovements()

5.3 Scoring Algorithm

The score engine in app.js uses a deterministic Linear Congruential Generator (LCG) seeded from a polynomial hash of the startup idea's input string. This ensures that the same idea always produces identical scores across sessions. Industry bonuses (AI/ML +8, SaaS +6, HealthTech +5, etc.) are applied on top. Each of the 5 category scores is clamped to [20, 99] using: *clamp(rand()×range + base + bonus, 20, 99)*. The overall score is the average of the 5 category scores.

MODULES AND FEATURES

IdeaProof produces 12 distinct output sections from a single idea submission. Each is documented below with its corresponding screenshot.

6.1 Landing Page and Idea Input Form

The landing page presents the IdeaProof brand with an "AI-Powered Validation" badge and the headline "Validate your startup idea *before you build it.*" Three feature pills — 20+ Signals, Free No signup, AI Powered — communicate the value proposition concisely. An Analyze My Idea CTA button smooth-scrolls to the form.

The input form (Step 1 — Your Idea) collects five fields: Startup Name (max 60 chars with live counter), Industry (dropdown, 13 categories), Idea Description (max 500 chars, min 20 required), Target Audience, and Region/Country. All fields are validated on submit before analysis begins.

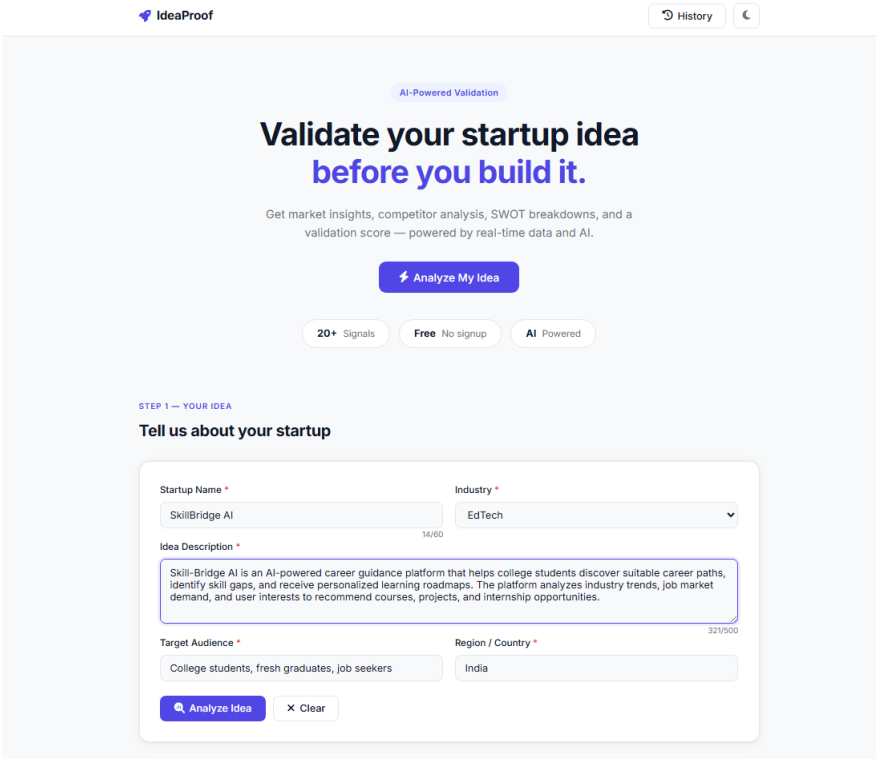


Figure 6.1 – Landing Page Hero and Idea Input Form (Step 1)

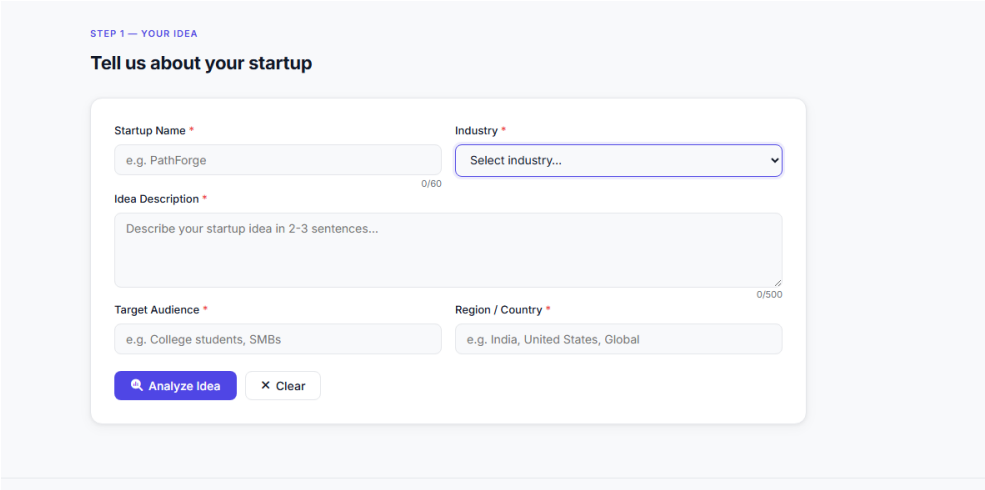


Figure 6.2 – Idea Input Form with Sample Data Entered

6.2 Validation Score and Score Breakdown

After analysis, the primary output is the Validation Score panel. An animated SVG arc gauge animates from 0 to the final score over 18ms intervals. The gauge colour reflects score quality: green ≥75, blue ≥55, amber ≥40, red below 40. A verdict badge (Great / Solid / Needs Refinement / Rethink) appears below the gauge.

The Score Breakdown panel shows five animated progress bars for: Market Need (base 50, range 35), Innovation (base 40, range 40), Scalability (base 45, range 35), Competition (base 35, range 40), and Revenue Potential (base 48, range 35) — each clamped to [20, 99].

The Startup Readiness Meter displays a 4-stage progression track: Idea Stage (0+), MVP Stage (40+), Launch Stage (60+), and Scale Stage (80+), with the current stage highlighted based on the overall score.

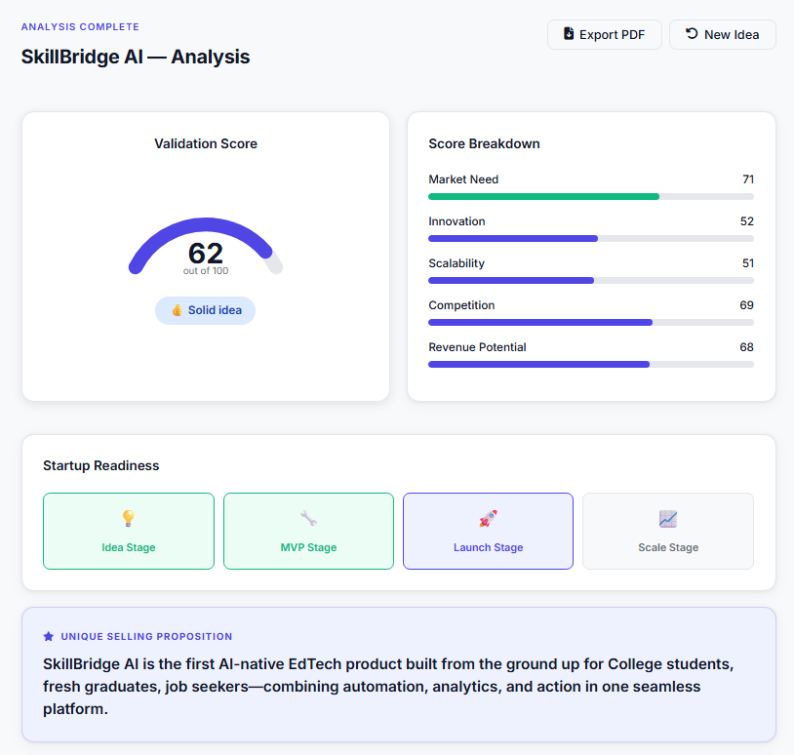


Figure 6.3 – Validation Score Gauge, Score Breakdown, and Startup Readiness Meter

6.3 SWOT Analysis

The SWOT Analysis section presents a four-quadrant grid generated by **generateSWOT()**. Each quadrant (Strengths, Weaknesses, Opportunities, Threats) is colour-coded — green, pink/red, blue, and amber respectively — and lists 3–4 bullet points selected from curated industry-aware pools using the seeded RNG, ensuring consistent results for the same idea across sessions.

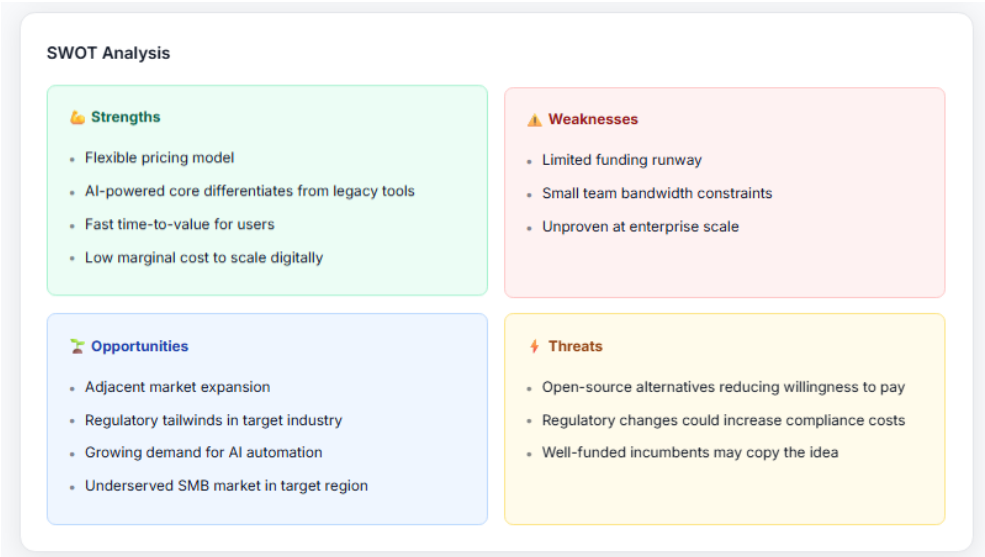


Figure 6.4 – SWOT Analysis with Four-Quadrant Grid

6.4 Competitor Analysis

The **fetchCompetitors()** function queries the Wikipedia API for companies in the startup's industry and the GitHub API for relevant repositories. Results are processed into competitor cards with name, description, market position badge (Leader / Challenger / Niche), establishment status, strengths, and a Visit link. A search bar and sort dropdown (by Name, Similarity, or Position) allow users to explore the competitor landscape interactively.

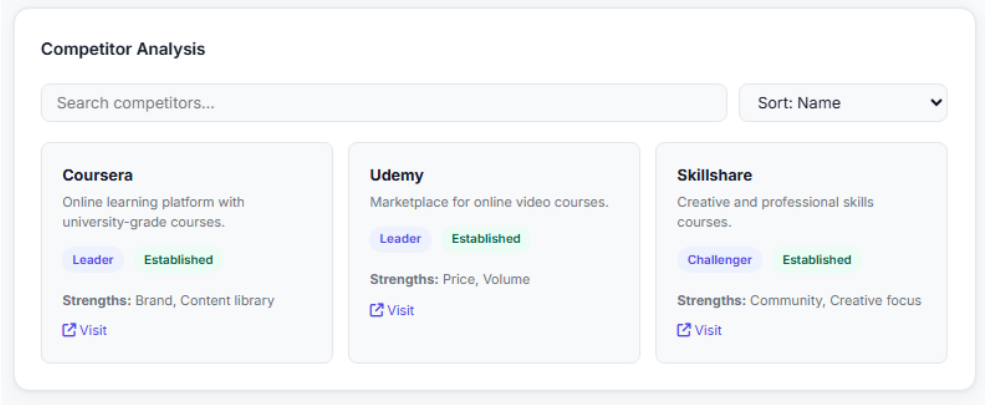


Figure 6.5 – Competitor Analysis Cards with Search and Sort

A Competitor Comparison Table below the cards presents all competitors in a structured format with Strengths, Weaknesses, Market Position, and a colour-coded Similarity percentage showing how closely each competitor overlaps with the user's idea.

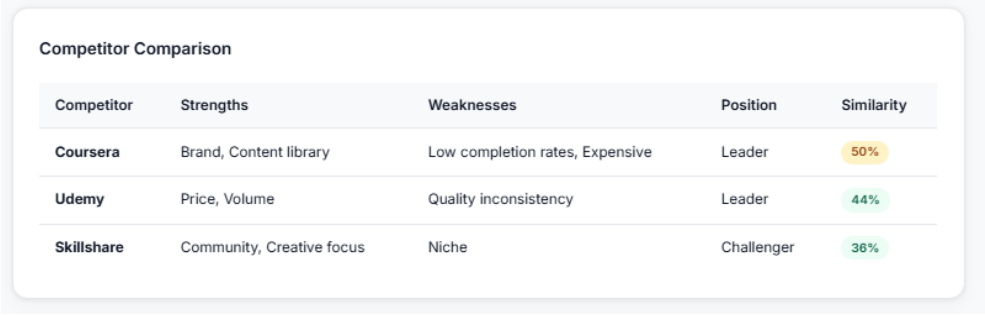


Figure 6.6 – Competitor Comparison Table with Similarity Scores

6.5 Revenue Model Suggestions

The **generateRevenue()** function evaluates 7 revenue models against the startup's industry and description, assigning a fit score to each. The highest-scoring model is tagged as **Top Pick**. Models include: Subscription, Affiliate, Freemium, Enterprise Licensing, Commission, Marketplace, and Advertising — each with a description, icon, and fit score out of 100.

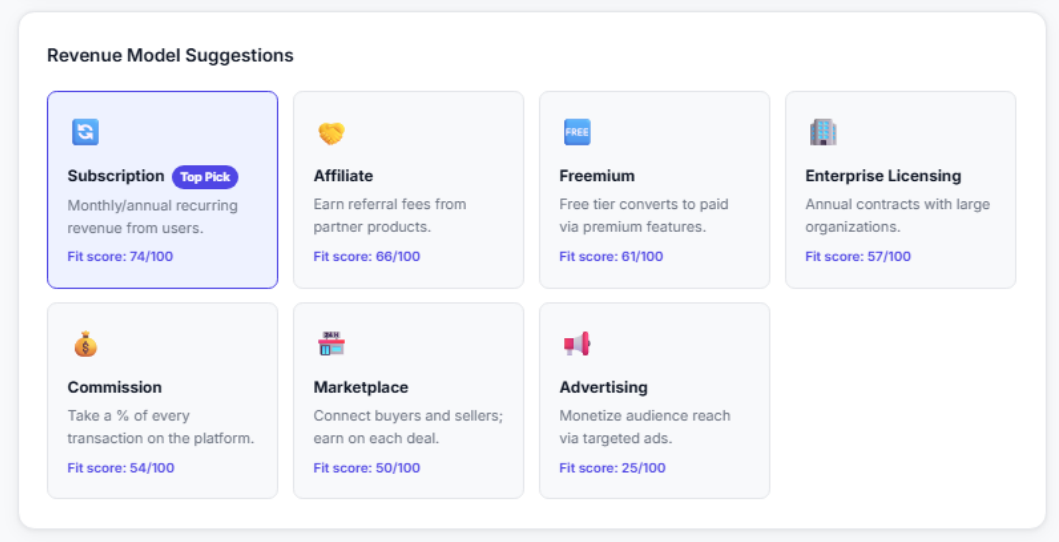


Figure 6.7 – Revenue Model Suggestions with Fit Scores and Top Pick

6.6 Target Audience Segmentation

The **generateAudience()** function produces three audience segment cards: Primary (the core early adopter group most acutely experiencing the problem), Secondary (organisations or teams that will adopt once ROI is proven), and Early Adopters (the first 100 users who value innovation and will provide critical feedback). Each card includes the segment type, audience name, and a strategic explanation of how they relate to the startup's growth journey.

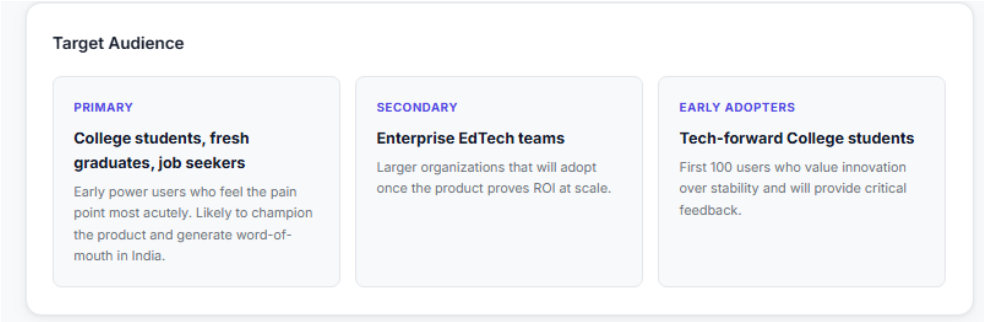


Figure 6.8 – Target Audience Segmentation: Primary, Secondary, and Early Adopters

6.7 Risk Assessment

The **generateRisks()** function produces 4 risk cards covering Competition, Technical, Financial, and Adoption risk categories. Each card is colour-coded by severity — amber for Medium Risk, red for High Risk, and green for Low Risk — with a description of the specific risk vector and a coloured severity bar.

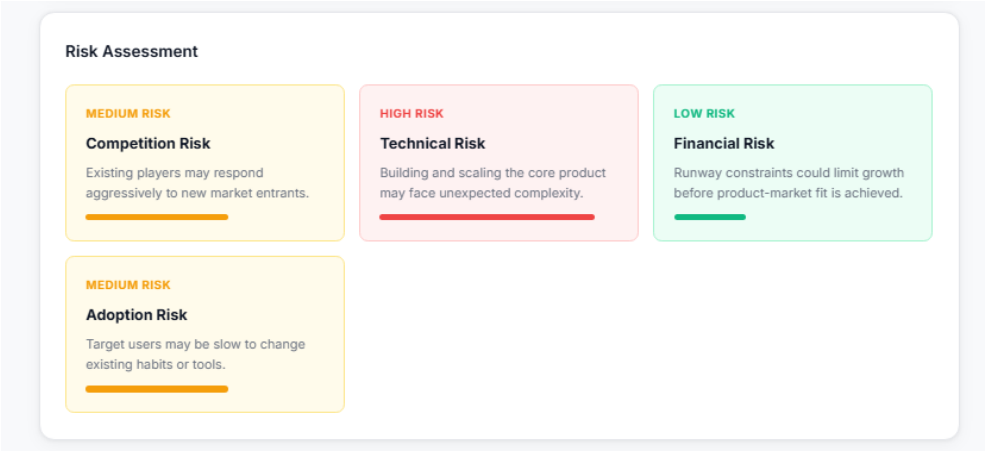


Figure 6.9 – Risk Assessment with Severity-Coded Risk Cards

6.8 Trend Analysis

The **generateTrends()** function produces two Chart.js visualisations: a bar chart showing signal strength across Market Demand, Investor Interest, Job Postings, Media Coverage, and Search Volume; and a line chart showing the metric's trajectory across 12 months. The **fetchLiveSignals()** function supplements this with real signal data from Hacker News and Reddit when available.

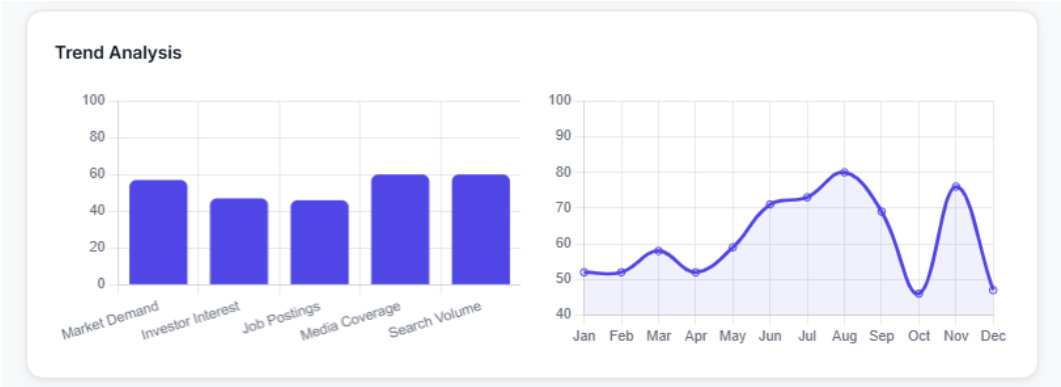


Figure 6.10 – Trend Analysis with Bar and Line Charts

6.9 Improvement Suggestions

The **generateImprovements()** function produces 10 numbered, actionable startup tips selected from a curated pool using the seeded RNG. Suggestions cover north-star metrics, waitlist validation, referral mechanics, community partnerships, audience narrowing, free trial strategy, customer advisory boards, content marketing, recurring revenue, and regional compliance — all tailored to the idea's industry and target region.

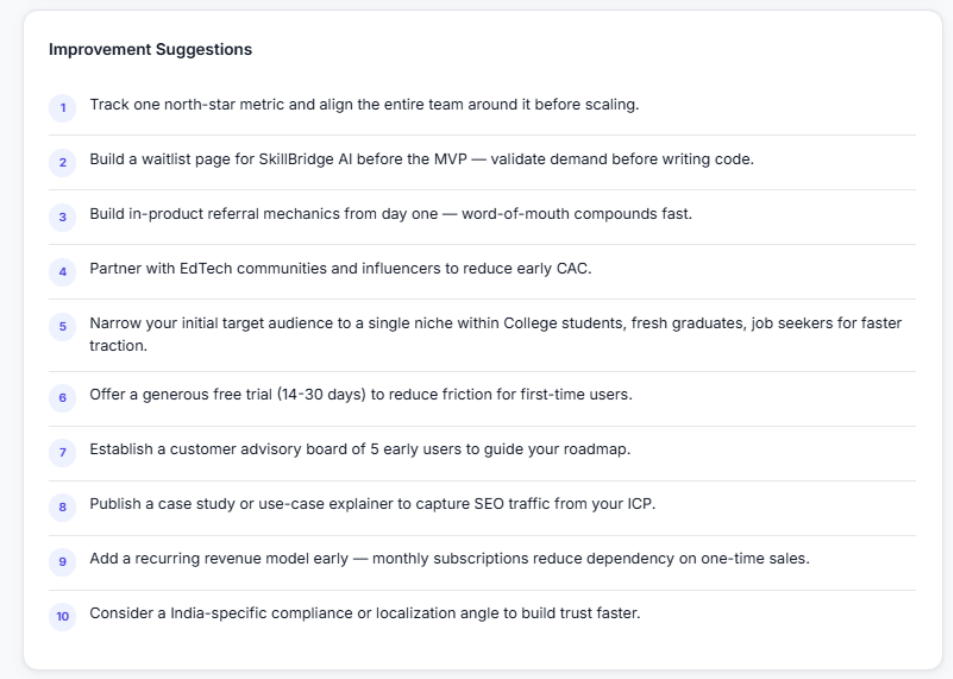


Figure 6.11 – Improvement Suggestions with 10 Numbered Actionable Tips

6.10 AI Startup Advisor Chatbot

The AI Startup Advisor is an interactive chatbot panel that allows founders to ask follow-up questions about their startup idea. Quick-question buttons — “How can I improve this idea?”, “Best revenue model?”, “Biggest risks?”, “Go-to-market strategy?” — pre-fill the input for rapid interaction. The **getAIAdvisorResponse()** function uses keyword matching against the idea context to generate relevant, specific responses. If a Groq API key is configured, it connects to a real LLM; otherwise it falls back to the **generateMockAdvisorResponse()** keyword-matched mock engine.

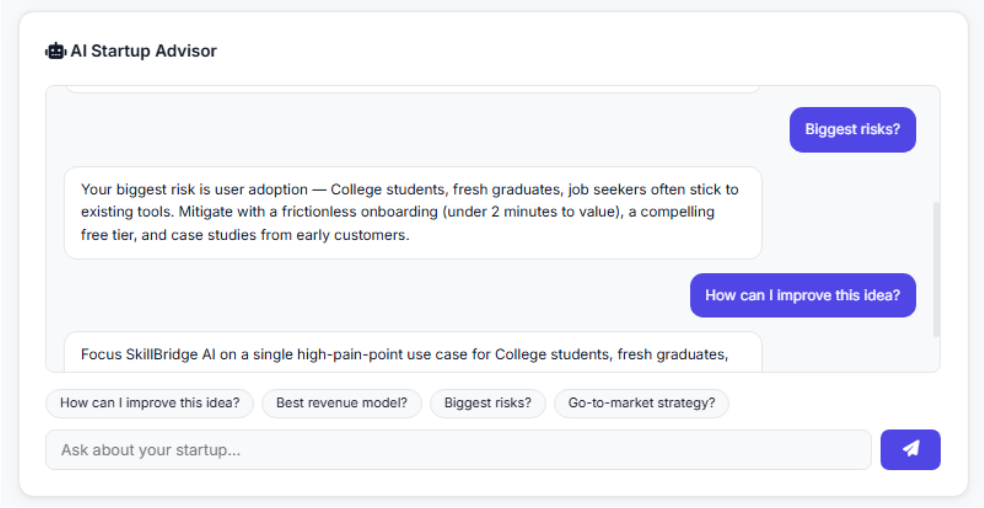


Figure 6.12 – AI Startup Advisor Chatbot with Quick Questions and Responses

TECHNOLOGIES USED

IdeaProof was built using only HTML5, CSS3, and Vanilla JavaScript, with four CDN-loaded libraries and four public APIs. No framework, no npm, no build step.

7.1 HTML5

HTML5 provides the complete structural markup for IdeaProof's single-page application in 259 lines. Semantic elements organise the navbar, hero section, idea input form, 7-step loading overlay, and all 12 results section containers. Form elements include a dropdown (13 industries), textarea with live character counter, and standard text inputs. CSS custom property references in HTML attributes enable the dark mode toggle.

7.2 CSS3

CSS3 delivers the complete visual system in 595 lines with zero hardcoded hex colours outside :root and [data-theme="dark"]. CSS Custom Properties power the dark mode toggle — switching a single [data-theme] attribute on the body element remaps all colours instantly. Components are styled individually (no utility framework). CSS keyframe animations handle the loading step indicators and score gauge animation. Responsive breakpoints at 700px and 420px ensure full mobile compatibility.

7.3 JavaScript (ES6+, Vanilla)

All 997 lines of app.js implement the complete application: form validation and submission handling, the **animateSteps()** async pipeline orchestrator, 10 generator functions using the seeded LCG random number generator, 4 API fetcher functions with try/catch fallback, 13 renderer functions writing to DOM containers, the Chart.js chart initialisation and data binding, the AI chatbot logic, localStorage history management, and the jsPDF export pipeline.

7.4 Chart.js

Chart.js renders two visualisations in the Trend Analysis section: a bar chart displaying market signal strengths across 5 dimensions, and a line chart showing the monthly trajectory of the key metric. Charts are initialised with data from **generateTrends()** and supplemented with real API signals from Hacker News and Reddit when available.

7.5 jsPDF

jsPDF (v2.5.1) generates the downloadable PDF business report. The **exportPDF()** function builds the document programmatically — adding a title page, startup metadata, validation score, category breakdown, SWOT summary, competitor list, revenue models, and improvement suggestions. The file

is downloaded as *{StartupName}-IdeaProof-Report.pdf*.

7.6 Public APIs (Wikipedia, GitHub, HN, Reddit)

Four free public APIs are integrated: the **Wikipedia API** is queried for the startup's industry to find real company articles for competitor cards; the **GitHub API** searches repositories related to the industry for additional competitor signals; the **Hacker News Algolia API** fetches recent stories matching the startup's keywords for live market signals; and the **Reddit JSON API** retrieves recent posts for community signal data. All four have graceful fallback to industry-specific mock datasets if they fail due to rate limits or CORS issues.

7.7 Netlify

IdeaProof is deployed as a static site on Netlify, providing HTTPS, a custom subdomain, and global CDN distribution for the three-file project.

#	Technology / Library	Type	Purpose
1	HTML5	Markup	SPA structure, form, results containers
2	CSS3 + Custom Props	Styling	Dark mode, responsive, animations
3	JavaScript ES6+	Programming	All logic, generators, renderers, chat
4	Chart.js	CDN Library	Bar + line trend charts
5	jsPDF 2.5.1	CDN Library	PDF report export
6	Font Awesome 6.5	CDN Library	UI icons throughout the interface
7	Google Inter	CDN Font	Typography
8	Wikipedia API	Public API	Real competitor data
9	GitHub API	Public API	Repository-based competitor signals
10	HN Algolia API	Public API	Live market trend signals
11	Reddit JSON API	Public API	Community discussion signals
12	Netlify	Platform	Static hosting and CDN deployment

CONCLUSION

IdeaProof is a technically ambitious and practically meaningful mini project that successfully replicates the core analytical output of a professional startup validation tool within a three-file, zero-dependency frontend application. By combining four public API integrations with a deterministic scoring algorithm, 10 generator functions, 13 renderer functions, interactive Chart.js visualisations, an AI chatbot, localStorage history, and PDF export — all in 1,851 lines of HTML, CSS, and Vanilla JavaScript — the project demonstrates exceptional technical scope for a student vacation exercise.

The most technically innovative aspect of IdeaProof is its deterministic seeded scoring engine. Using a Linear Congruential Generator seeded from a polynomial hash of the input string ensures that identical ideas always produce identical analysis results — a property essential for a validation tool where reproducibility matters. This approach allows the application to behave like a consistent analytical engine rather than a random output generator, significantly enhancing the credibility of the results.

The API integration architecture, with graceful fallback to industry-specific mock datasets when any of the four APIs are unavailable, demonstrates mature software design thinking. The application degrades gracefully rather than breaking, ensuring a consistent user experience regardless of network conditions, API rate limits, or CORS restrictions.

The breadth of the output — 12 distinct analysis sections covering scoring, SWOT, competitors, revenue models, audience, risks, trends, improvements, and an AI advisor — provides genuine business utility that goes well beyond what is typically expected of a student project. A founder could realistically use IdeaProof as a first-pass validation tool before committing to deeper market research.

On a personal level, building IdeaProof independently during the vacation period has been a comprehensive exercise in full-application JavaScript development. Designing the scoring algorithm, orchestrating the async 7-step pipeline, integrating four APIs with fallback logic, and building the chatbot response engine provided exposure to software architecture concepts that extend well beyond standard coursework.

In conclusion, IdeaProof is a polished, feature-complete, and publicly deployed startup validation platform that fully achieves its stated objectives and stands as one of the most technically comprehensive mini projects produced during this academic period.

Live Project: https://startup-validator-bhoomika.netlify.app/